Module 6: Introduction to Bootstrap

Wenting Xu

Iowa State University

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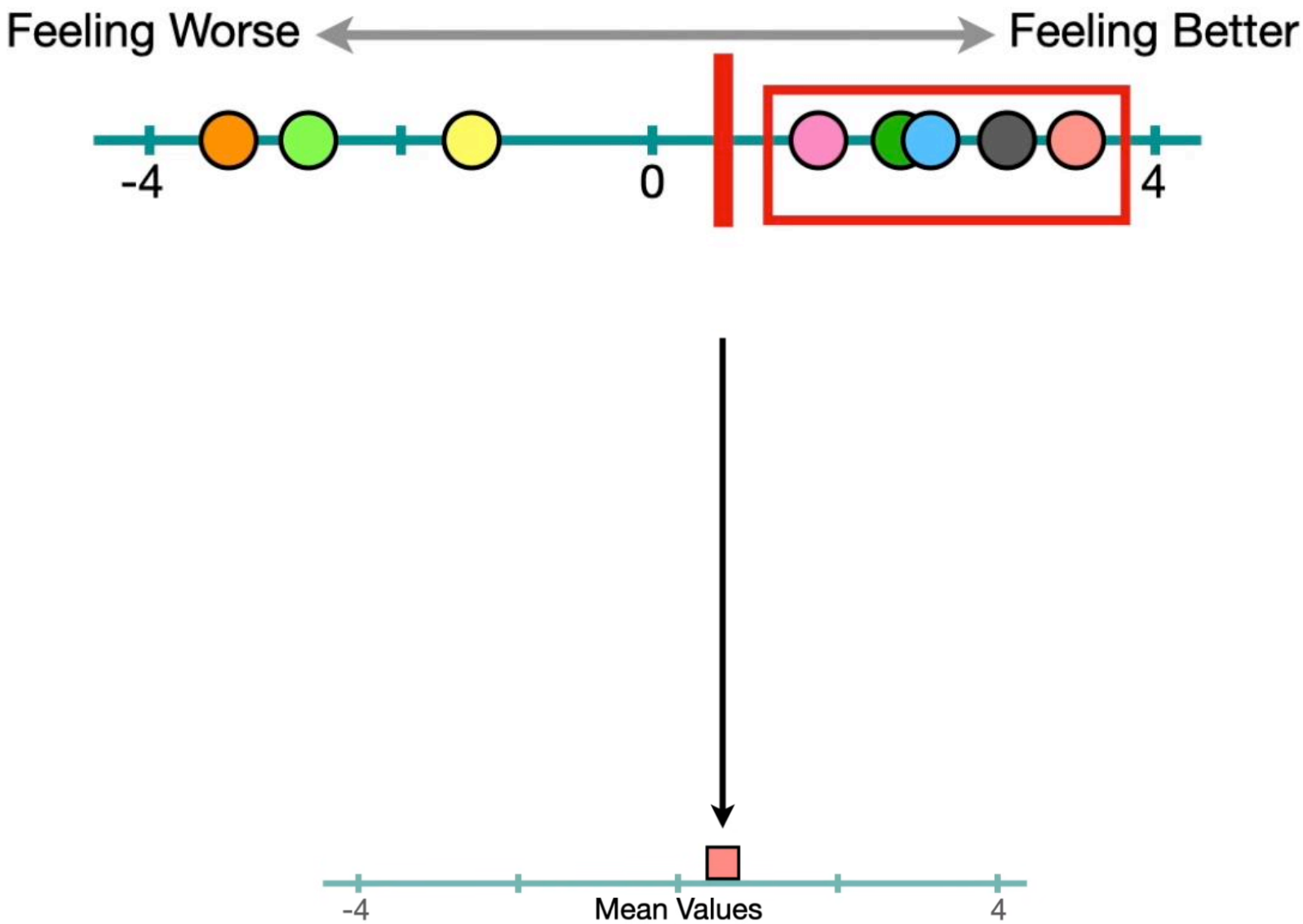
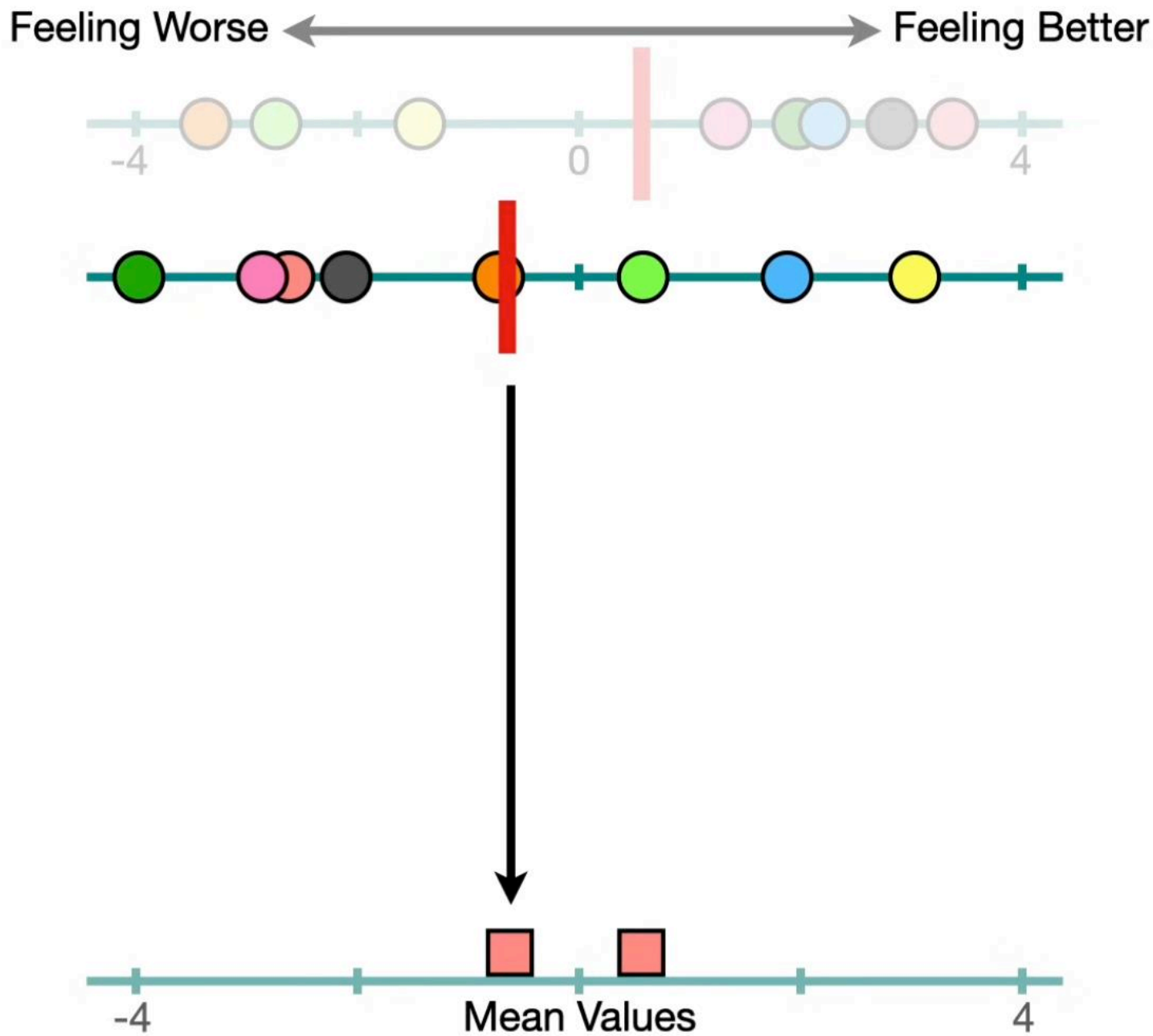
Some lecture slides and instructional materials in this course are adapted from the following sources:

- Website: https://statquest.org/index.html


Feeling Worse
Feeling Better
-4
0
4
-4
Mean Values
4


Feeling Worse
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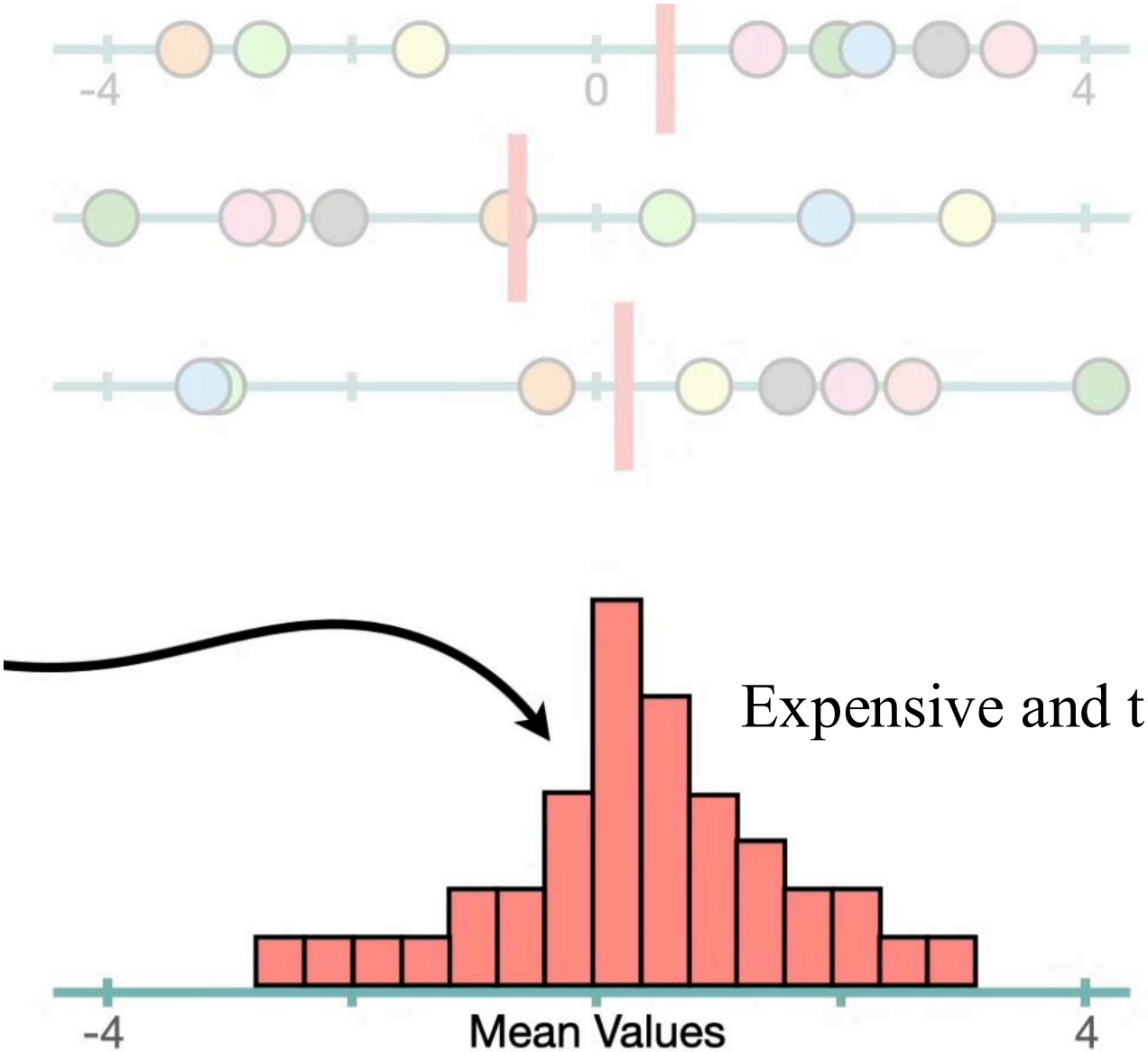
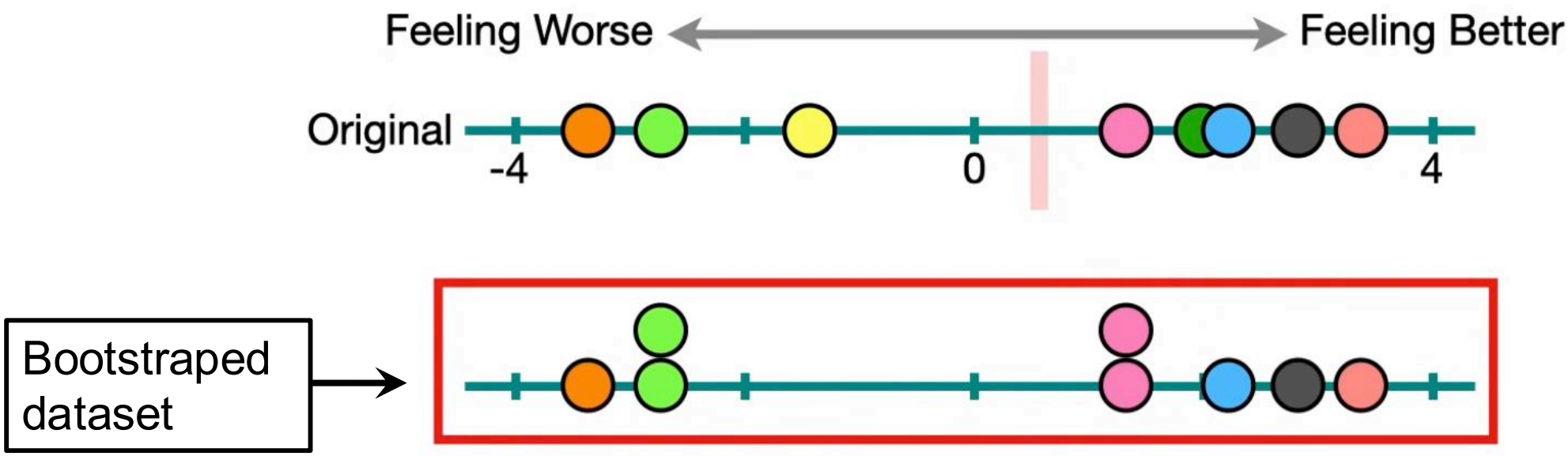
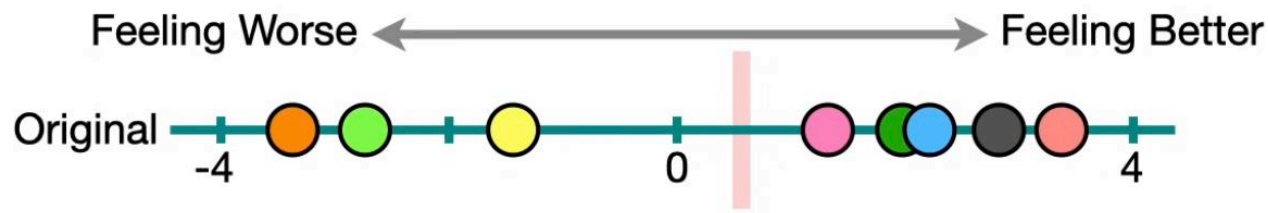


Expensive and time consuming



Sampling with replacement: randomly selecting observations from the dataset, where the same observation can be chosen more than once.



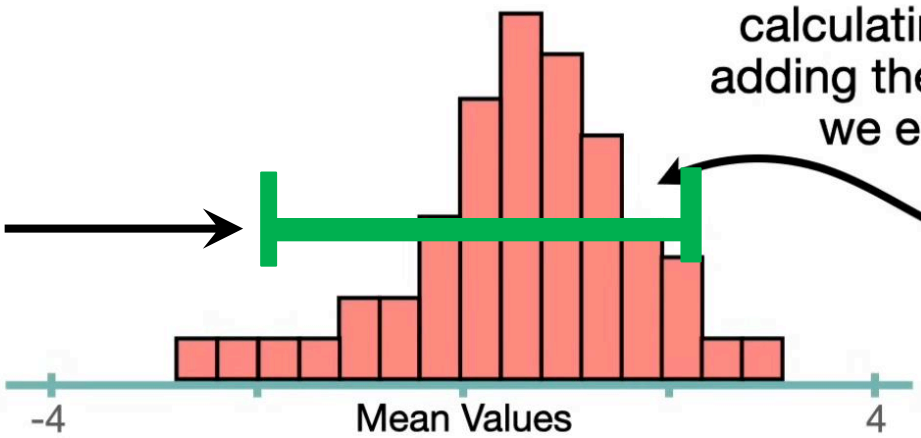
Bootstraped dataset 1

Bootstraped dataset 2

. . . . . .

…and, after creating **1,000s** of **Bootstrapped Samples**, calculating their means and adding them to the histogram, we end up with this.

95% Confidence Interval



Standard error of sample mean: unknown

Standard deviation of bootstraped means

Resampling Techniques

1. Cross-validation:
   - Used to estimate supervised test error (prediction or classification error)
   - Can also help us to find an optimal tuning parameter (such as λ in regularized regression)
2. Bootstrap
   - Used to estimate uncertainty surrounding a statistical approach.
   - Common examples:
     - Estimate the standard error of parameter estimates
     - Construct confidence intervals.

Bootstrap

- One of the most important techniques in all of data science/statistics.
- Widely applicable, extremely powerful, computationally intensive.
- Literally involves just resampling from the data. It lets us mimic what would happen if we collected new data over and over again, without actually collecting new data.
- No distributional assumptions.
- Requires a moderately sized data set. $(n > 35)$

sample size

2

Recall `lm()` output

```
> summary(lm(medv~.,data=Boston))

Call:
lm(formula = medv ~ ., data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.595  -2.730  -0.518   1.777  26.199 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.646e+01  5.103e+00   7.144 3.28e-12 ***
crim        -1.080e-01  3.286e-02  -3.287 0.001087 ** 
zn           4.642e-02  1.373e-02   3.382 0.000778 ***
indus        2.056e-02  6.150e-02   0.334 0.738288    
chas         2.687e+00  8.616e-01   3.118 0.001925 ** 
nox         -1.777e+01  3.820e+00  -4.651 4.25e-06 ***
rm           3.810e+00  4.179e-01   9.116  < 2e-16 ***
age          6.922e-04  1.321e-02   0.052 0.958229    
dis         -1.476e+00  1.995e-01  -7.398 6.01e-13 ***
rad          3.060e-01  6.635e-02   4.613 5.07e-06 ***
tax         -1.233e-02  3.760e-03  -3.280 0.001112 ** 
ptratio     -9.527e-01  1.308e-01  -7.283 1.31e-12 ***
black        9.312e-03  2.686e-03   3.467 0.000573 ***
lstat       -5.248e-01  5.072e-02 -10.347  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

$\hat{\beta}$ $SE(\hat{\beta})$

How did we obtain these standard errors and carry out inference?

Inference = Hypothesis Testing, Confidence Interval

$SE(\hat{\beta})$

assumptions:

↳ constant variance $[\sigma^2]$

↳ we have a good estimate of σ^2 $[\hat{\sigma}^2]$

↳ error term follows normal distribution

How did we obtain these standard errors and carry out inference?

What happens if

- The modeling assumptions break down?
- There is no analytical formula that can be derived.

Bootstrap standard error and confidence intervals!

- Requires no math.
- Requires no distributional assumptions.

Bootstrap standard errors

Suppose we are working on the lasso model. We estimate our parameters $\hat{\beta}_1$, $\hat{\beta}_2$, …., $\hat{\beta}_p$

We want to quantify: how accurate are our estimates?

1) resample from the original distribution
2) get many $\hat{\beta}_1$'s
3) calculate the standard error

Bootstrap standard errors

B = 1000
for b = 1,2,….B:

1. draw a bootstrap sample: bootstrap sample b
2. With this bootstrap sample, calculate a statistic $\hat{\beta}_1^{(b)}$

$\hat{\beta}_1^{(1)}, \hat{\beta}_1^{(2)}, \ldots\ldots\ldots, \hat{\beta}_1^{(B)}$

$$\mathrm{SE}(\hat{B}_1) = \sqrt{\frac{\sum_{b=1}^{B}\left[\hat{\beta}_1(b) - \bar{\beta}_1\right]^2}{B-1}}$$

- This can be applied to any parameter θ, and its estimate $\hat{\theta}$.

Bootstrap confidence intervals

- Classically, confidence intervals require distributional assumptions.
- Recall from linear regression, if I wanted to construct a confidence interval for $\hat{\beta}_1$ I needed one of the following:

Assumptions:

↳ $\varepsilon \sim N(0, \sigma^2)$

↳ central limit theorem

↳ n is large enough

↳ $n \to \infty$, $\hat{\beta} \to$ normal distribution

Bootstrap confidence interval

B = 1000
(Outer loop)
for b = 1,2,….B:

1. draw a bootstrap sample: call it bootstrap sample (b)
2. With this bootstrap sample, compute $\hat{B}_1^{(b)}$
   (Inner loop)
   For m = 1,2, …, M (M = 200):
   1) draw a bootstrap sample: bootstrap sample (b,m)
   2) compute $\hat{B}_1^{(b,m)}$

   $SE(\hat{B}_1^{(b,m)}) = \ldots.\ (\underbrace{\hat{B}_1^{(b,1)},\ \hat{B}_1^{(b,2)},\ \ldots\ldots\ \hat{B}_1^{(b,M)}}_{200})$

inside outer loop

$t_{boot,b} = \dfrac{\hat{B}_1^{(b)} - \hat{B}_1}{SE(\hat{B}_1^{(b)})}$ ← original data (before bootstrapping)